Measuring Neuromuscular Electrophysiological Activities to Decode HD-sEMG Biometrics for Cross-Application Discrepant Personal Identification With Unknown Identities

Xinyu Jiang, Xiangyu Liu, *Member, IEEE*, Jiahao Fan, Xinming Ye, Chenyun Dai, *Member, IEEE*, Edward A. Clancy, *Senior Member, IEEE*, and Wei Chen, *Senior Member, IEEE*

***Abstract*— Measuring the physical, physiological, behavioral, or chemical characteristics of an individual as biometrics for personal identification has attracted increasing attention in smart environment applications. Noncancelability and cross-application invariance are two flaws of traditional DNA, face, and fingerprint-based biometrics because users cannot volitionally change the biometric template. In this work, we acquired high-density surface electromyogram (HD-sEMG) signals encoded by gesture passwords as biometrics. The different sEMG patterns under different motor tasks allow users to enroll multiple accounts using sEMG under different hand gestures as biometrics. By simply changing to a new gesture password, users can cancel the original template once it is compromised. Even if impostors enter the correct gesture password, the individual differences of HD-sEMG as the second defense can still achieve excellent performance. To improve the current state-of-the-art identification accuracy, we acquired 256-channel forearm HD-sEMG and decoded high-resolution neuromuscular information in temporal–spectral–spatial domain. We achieved a high identification accuracy of 99.85% on a 200-account (20 subjects × 10 accounts per subject) recognition task, with training and testing data acquired 3 to 25 days apart. Moreover, to address the concern of "unknown identities," we applied an "authentication + identification" validation, achieving high accuracy of 93.81% on a 200-account [(16 enrolled subjects + 4 unknown subjects) × 10 accounts per subject] task. Our work substantially improves the current state-of-the-art accuracy for cross-day sEMG biometric identification (improved from ∼88% to >99% with a similar number of identified classes).**

***Index Terms*— Biometrics, high-density surface electromyogram (HD-sEMG), user identification.**

Manuscript received 27 January 2022; revised 29 April 2022; accepted 25 May 2022. Date of publication 6 June 2022; date of current version 27 October 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62173094 and in part by the Shanghai Municipal Science and Technology Project under Grant 20510710500. The Associate Editor coordinating the review process was Jagadeesh Varadarajan Kumar. *(Corresponding authors: Chenyun Dai; Wei Chen.)*

Xinyu Jiang and Chenyun Dai are with the Center for Intelligent Medical Electronics, School of Information Science and Technology, Fudan University, Shanghai 200433, China (e-mail: chenyundai@fudan.edu.cn).

Xiangyu Liu is with the College of Communication and Art Design, University of Shanghai for Science and Technology, Shanghai 200093, China.

Jiahao Fan and Wei Chen are with the Center for Intelligent Medical Electronics, School of Information Science and Technology, Fudan University, Shanghai 200433, China, and also with the Human Phenome Institute and the Zhangjiang Fudan International Innovation Center, Fudan University, Shanghai 201210, China (e-mail: w_chen@fudan.edu.cn).

Xinming Ye is with the School of Sports Science and Engineering, East China University of Science and Technology, Shanghai 200237, China.

Edward A. Clancy is with the Department of Electrical and Computer Engineering, Worcester Polytechnic Institute, Worcester, MA 01609 USA.

Digital Object Identifier 10.1109/TIM.2022.3180434

I. Introduction

THE rapid development of body sensor network (BSN)-based smart environment contributes to an increasing demand for personal identification to avoid privacy disclosure [1]. Measuring the physical, physiological, behavioral, or chemical characteristics of an individual as biometrics to recognize the identity of users is the most advanced and convenient solution. However, most biometric tokens (e.g., face and fingerprint) are easy to expose. Furthermore, the noncancelability nature of most biometrics (e.g., DNA [2], face [3], [4], fingerprint [5], and palmprint [6]) means that the biometric template is compromised forever once exposed because the template cannot be volitionally replaced. Moreover, if the same biometric token is used in multiple applications, the exposed biometric token in one application threatens the others. In more general application scenarios, a user may need to register multiple different accounts to switch between modes in a specific application or roles in a specific scenario. The cross-application invariance of traditional biometrics limits such scenarios.

The surface electromyogram (sEMG), with sufficient individually unique spinal nerve information, has emerged as a promising biometric trait and received increasing attention recently [7]. Furthermore, sEMG has been used in a very wide range of BSN-based smart healthcare and smart environment applications. For example, wearable sEMG measurement has been used in a physiological monitoring system for upper limb rehabilitation training [8]. In more general daily life applications, sEMG-based user interfaces are employed to manipulate a mobile phone [9]. With the wide applications of wearable sEMG in our daily life, measuring sEMG for personal identification in BSN applications is a convenient and efficient approach. Our previous work [7] demonstrated the promising cancelability of high-density sEMG (HD-sEMG) biometrics in user authentication applications for the first time.

1557-9662 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

A follow-up work [10] validated the cancelability and cross-application discrepancy of HD-sEMG biometrics in personal identification applications. In these previous studies [7], [10], HD-sEMG acquired from the right dorsal hand during *isometric* contractions of muscles corresponding to different fingers was used as biometric tokens. However, isometric contraction, despite its difficulty in being observed by impostors, is not a natural or comfortable manner to generate sEMG signals compared with dynamic contraction. EMG-based biometrics under different hand motions or gestures with dynamic contractions have been preliminarily investigated in previous studies [11], [12]. However, the training and testing data used in these previous studies were acquired on the same day. Accordingly, these studies did not take signal characteristic variation across days into consideration. Our previous work [13] investigated the cross-day performance of HD-sEMG biometrics under different dynamic hand gestures. However, only personal *authentication* was investigated. Compared with the personal authentication task that gives a binary decision of "yes" or "no" to verify if an individual person matches a specific user, personal *identification* is much more challenging, in which it seeks the identity label of the person in an enrolled database. The existing evidence of high identification accuracy of HD-sEMG biometrics in a cross-day validation is far too sparse to support practical use. Moreover, almost all previous studies on sEMG biometrics investigated personal authentication and identification as two separate problems. However, in practical personal identification applications, the system may face the risk of "unknown identities." Therefore, an identification system should be capable of first authenticating if the target person has already registered in the system before giving the identity label of the target person. In this case, "authentication + identification" should be properly integrated. Furthermore, the concerns of "unknown identities" (e.g., unregistered persons) have not been systematically investigated in previous studies.

In this work, we aim to improve the current state-of-the-art in HD-sEMG biometric identification. We acquired 256-channel forearm HD-sEMG under 34 hand gestures. Although HD-sEMG has been widely employed in other neural interface applications [14], our work is currently the only study measuring the neuromuscular electrophysiological activities from the whole forearm to decode sufficient information by using large-number and HD electrodes for the purpose of biometric identification. HD-sEMG acquired from the forearm, compared with dorsal hand sEMG used in our previous work [7], [10], contains more muscular and spinal nerve information because muscle groups and motor units functioning in the hand and wrist are anatomically located in the forearm. The dorsal hand location is limited to intrinsic hand muscles only. For physiological measurement, selecting different measurement locations and ranges will normally lead to a greatly different outcome. Temporal–spectral–spatial features were extracted from sEMG to represent the biometric template. Specifically, the variance of central frequency (VCF) and skewness features were extracted as representations of HD-sEMG biometrics. VCF feature was first applied in sEMG biometrics studies in our previous work [13]. In this work, we first propose to apply skewness, a simple but powerful statistical parameter of sEMG signals, as an effective representation of sEMG biometrics. Results show that skewness is a powerful supplementary representation of the VCF feature. The identification accuracy was evaluated in a cross-day validation protocol with training and testing data acquired 3 to 25 days apart. To further validate the cancelability and cross-application discrepancy of HD-sEMG biometrics, we enrolled multiple accounts for each person. Our work achieved an accuracy of 99.96% and 99.85% for 40-account (20 subjects $\times$ 2 accounts per subject) and 200-account (20 subjects $\times$ 10 accounts per subject) recognition tasks, respectively. Even if all subjects entered the same gesture password, the individual differences in HD-sEMG characteristics still achieved a high identification accuracy of 96.60%. Moreover, to address the concern of "unknown identities," we applied an "authentication + identification" validation procedure, achieving high accuracy of 93.81% on a 200-account [(16 enrolled subjects + 4 unknown subjects) $\times$ 10 accounts per subject] task. We also investigated the variation of password entropy of the HD-sEMG biometric template with the increasing length of selected gesture passwords to evaluate the security of HD-sEMG biometrics against the brutal attack. We found an approximately linear growth of template entropy with the increasing length of gesture passwords, indicating the relative independence of information provided by different gestures. The obtained accuracy substantially improves the state-of-the-art performance of sEMG-based personal identification (improved from the level of $\sim$88% to $>$99%, with a similar number of identified classes), representing the new state-of-the-art accuracy.

Overall, the contributions of our work are summarized in the following.

1) Evaluating the performance of sEMG biometrics in a rigorous cross-day validation, with an accuracy of $>$99% obtained, which is currently the highest cross-day identification accuracy in the literature (with considerably substantial improvement from the previous level of $\sim$88%).
2) Evaluating the impact of unknown identities on sEMG biometric identification, which is ignored by previous studies but certainly a crucial factor in practical applications.
3) Analyzing the improvement of identification accuracy achieved in this work from the technical aspect. Other than the VCF feature (first applied in sEMG biometric identification in our previous work [13]), we introduce skewness as an sEMG biometric representation. By comparing the performance of a total of 19 features used in our and previous studies [7], [10]–[13], [15]–[17], we found that VCF is the most powerful single feature for sEMG representation. Skewness is the optimal supplementary representation of sEMG combined with VCF with the highest identification accuracy achieved using the combination of VCF and skewness.
4) Analyzing the improvement of identification accuracy achieved in this work from the data sensing aspect. We measured the HD-sEMG biometrics from the whole

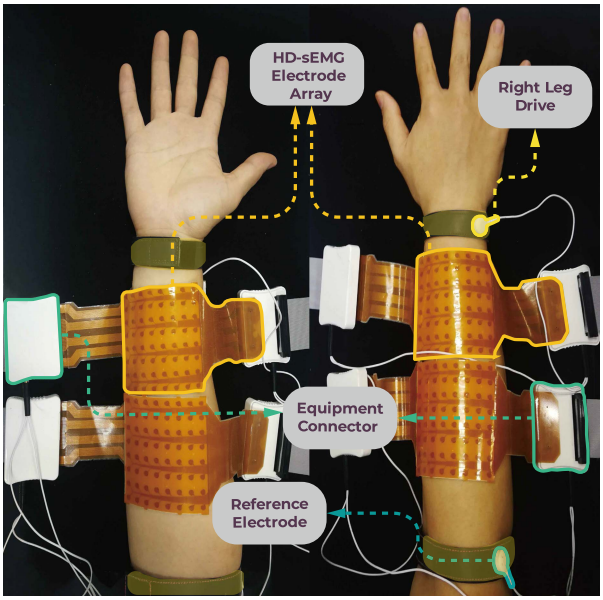


Fig. 1. Setup of the electrode arrays.

forearm with 256 electrodes. We further proved that a large area covered by electrodes, rather than the total number of electrodes, is a key factor for better identification performance. We also found that, with the same number of electrodes, a larger area covered by sparsely distributed electrodes is more effective than densely distributed electrodes within a small area.

5) Finding an approximately linear growth of entropy of sEMG biometric templates with the increasing length of gesture password, indicating the relative independence of information provided by different gestures. With increasing hand gestures, the security of sEMG biometrics against brutal attacks grows exponentially (evidenced by the linear growth of entropy).

II. Materials

A. Subjects

Twenty subjects (eight females and twelve males, age: 21–34 years old) were recruited for our data acquisition experiment. We informed all recruited subjects about the experiment protocol and research purpose. All subjects provided written informed consent. The experiment protocol was approved by the ethics committee of Fudan University (approval number: BE2035). Our data (Hyser dataset) were open-sourced in our previous work [18]. Here, we only briefly introduce the dataset. For more details, please refer to [18].

B. Data Acquisition

Four electrode arrays with 8 × 8 electrode layout of each array (256 channels totally) were employed to measure the forearm HD-sEMG, as shown in Fig. 1. The reference electrode and the right leg drive were fixed on the elbow and the head of the ulna, respectively. Quattrocento amplifier (OT Bioelettronica, Turin, Italy) was used in HD-sEMG measurement, with a gain of 150, a 16-bit resolution, a 2048-Hz sampling rate, and a 10–500-Hz passband.

Fig. 2. All gestures performed by subjects.

During the experiment, subjects followed experiment instructions presented on the screen to complete the experimental tasks. Subjects were asked to perform 34 different hand gestures at their most comfortable effort levels. These tasks are itemized and shown in Fig. 2. Subjects performed two successive trials for each gesture before continuing to the next one, following the order listed in Fig. 2. Fig. 3 presents the experimental protocol in each trial. Three 1-s dynamic tasks (motions from relaxing to the target gestures) and one 4-s maintenance task (motions from relaxing to the target gestures followed by maintenance at the required gestures) were performed in a single trial. Subjects were provided a 2-s resting duration between two successive tasks in each trial and a 5-s resting duration between two successive trials. At both the onset and the end of each task, the experiment instruction software sent a trigger signal to an unused channel of the acquired data (the 257th channel) to synchronize the acquired data and the experiment. A total of 204 dynamic tasks (34 gestures × 2 trials × 3 dynamic tasks) and 68 maintenance tasks (34 gestures × 2 trials × 1 maintenance task) were performed by each subject in each experiment. Subjects informed the experiment assistant if they performed a task incorrectly or missed a specific task. Such tasks were excluded from the final dataset for analysis. An average of 1.13% ± 1.33% dynamic tasks and 1.25% ± 1.54% maintenance tasks were excluded. We measured the HD-sEMG of each subject twice in two sessions on separate days (3 to 25 days apart, 8.50 ± 6.72 days

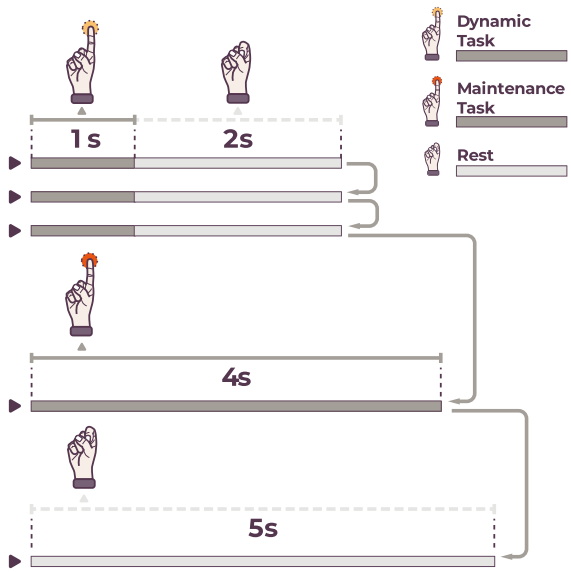


Fig. 3. Required tasks in a single trial.

on average), following the same experimental protocol as described above. Data acquired in sessions 1 and 2 were allocated as the training and testing datasets, respectively.

The acquired data were analyzed using MATLAB R2020b (MathWorks, Inc.) on an Intel[1] Xeon[2] processor (CPU E5-1630 v4 @ 2.20 GHz).

III. Methods

A. Data Preprocessing

The acquired 256-channel HD-sEMG signals were filtered with 10-Hz high-pass and 500-Hz low-pass Butterworth filters. Both the high- and low-pass filters were zero-phase digital filters, which processed the input time series bidirectionally, eight-order each direction). Notch filters were applied to suppress the power line interference at 50 Hz and all harmonics up to and including 400 Hz. The preprocessed HD-sEMG signals were then segmented into different tasks. The first 0.25-s (reaction period of subjects) signal of each task was removed. Accordingly, signals with 0.75- and 3.75-s durations were used for further analysis for dynamic and maintenance tasks, respectively.

B. Feature Extraction

VCF and skewness features were extracted to represent the biometric template in the temporal–spectral–spatial domain. The details of the employed features were introduced in the following.

1) VCF Feature: VCF is an important indicator of the power spectral density (PSD) of sEMG. To calculate VCF, the spectral moments (SMs) need to be calculated first. The zeroth SM and first SM (denoted by SM0 and SM1, respectively) can be obtained by

$$\mathrm{SM0} = \sum_{k=0}^{K} P_k \tag{1}$$

$$\mathrm{SM1} = \sum_{k=0}^{K} P_k f_k \tag{2}$$

where f_k is the frequency of the sEMG spectrum at frequency bin k, P_k is the power at f_k, and K is the length of the frequency bin ($K = 513$ in our work with a 2-Hz resolution from 0 to 1024 Hz). To obtain P_k at each f_k, Welch's averaged, the modified periodogram method was applied to estimate the PSD using a 1024-length Hamming window and 50% overlap. Then, the central frequency of sEMG PSD (denoted by f_c) can be calculated by

$$f_c = \frac{\sum_{k=0}^{K} P_k f_k}{\sum_{k=0}^{K} P_k} = \frac{\mathrm{SM1}}{\mathrm{SM0}}. \tag{3}$$

We further define the second SM (denoted by SM2) as

$$\mathrm{SM2} = \sum_{k=0}^{K} P_k f_k^2. \tag{4}$$

The VCF can be calculated by

$$\mathrm{VCF} = \frac{1}{\mathrm{SM0}} \sum_{k=0}^{K} P_k (f_k - f_c)^2 = \frac{\mathrm{SM2}}{\mathrm{SM0}} - \left(\frac{\mathrm{SM1}}{\mathrm{SM0}}\right)^2. \tag{5}$$

2) Skewness Feature: Skewness is an asymmetry measure of the probability distribution of sEMG signal values in the temporal domain, defined as

$$\mathrm{Skewness} = \frac{\sum_{t=1}^{T} (x_t - \bar{x})^3}{\left(\sum_{t=1}^{T} (x_t - \bar{x})^2\right)^{\frac{3}{2}}} \tag{6}$$

where x_t is the tth sampling value, T is the total number of the sampling values, and $\bar{x}$ is the mean of the sEMG segment (0.75- and 3.75-s duration segments for dynamic and maintenance tasks, respectively).

Each feature (VCF or skewness) was computed once in each channel for each task, yielding a 256-length feature vector. Each element in the 256-length feature vector was obtained from the sEMG signal in a specific channel (a total of 256 sEMG channels).

Although HD-sEMG provides muscular information with high spatial resolution, outliers in "bad" channels can greatly reduce identification accuracy. Therefore, we detected outliers within each individual feature if a value was more than three standard deviations away from the mean value of the ensemble for that respective feature. The ensemble included 64 feature values extracted from each 8×8 electrode array from each task. Outlier detection was performed separately for each individual task. The detected outliers were then smoothed by

[1]Registered trademark.
[2]Trademarked.

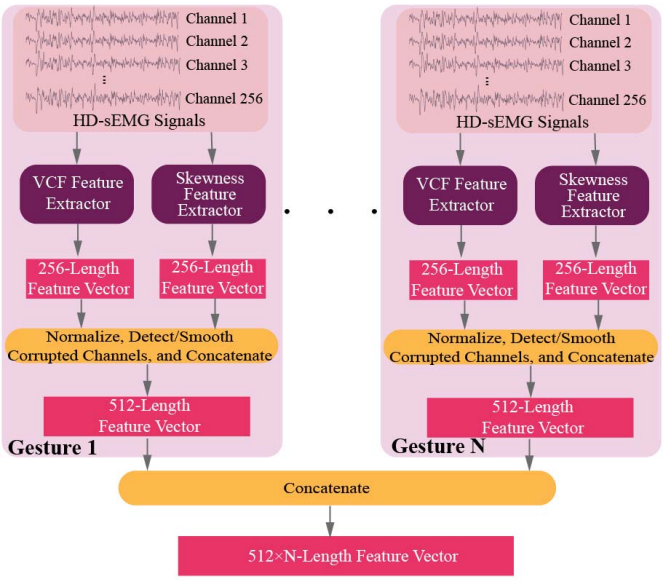


Fig. 4. Structure of feature vectors.

the average of features in their four neighbor channels (two or three neighbor channels for the vertex or edge channels of the electrode array).

Each 256-length feature vector (one for each of the two features, VCF or skewness) was normalized to have a mean value of zero and a standard deviation of one. These two (VCF and skewness) normalized vectors were then concatenated into one combined feature vector of length 512 (for each gesture). If multiple gestures were performed to enter a gesture password (a sequence of gestures), the feature vectors corresponding to each gesture were concatenated to construct a $512 \times N$-length feature vector (where N is the number of gestures in the gesture password). The structure of feature vectors is presented in Fig. 4. The linear support vector machine (SVM) classifier ("one versus all" strategy for the investigated multiclass classification) was employed to evaluate the identification accuracy of the proposed method using the combined feature vectors with data acquired from the first and second days as training and testing sets, respectively.

C. EMG Features Used as Biometrics Representations in Previous Studies

Besides VCF and skewness features, we also extracted 17 other features used in previous sEMG biometrics studies. We demonstrated the superiority of our features by comparing with the others. These features used in previous sEMG biometrics studies were: 1) root mean square (rms) [7], [12], [13], [15]–[17]; 2) waveform length (WL) [7], [10]–[12], [15]–[17]; 3) sample entropy (SampEn) [7], [17]; 4) spectral entropy (SpEn) [7], [13], [17]; 5) median frequency (MDF) [7], [10], [17]; 6) difference absolute standard deviation value (DASDV) [15]; 7) mean absolute value (MAV) [11], [15], [16] (in [16], this feature was termed integral EMG (IEMG), which is 100% correlated with MAV); 8) simple square integral (SSI) [16]; 9) Hjorth1 parameter [15], [16] (Hjorth1 is 100% correlated with variance of the input data); 10) Hjorth2 parameter [13]; 11) zero crossing (ZC) [11], [12], [15]; 12) slope sign change (SSC) [11], [12]; 13) log detector (LOGD) [15]; 14) third-order absolute temporal moment (TM3) [15]; 15) fourth-order absolute temporal moment (TM4) [15]; 16) fifth-order absolute temporal moment (TM5) [15]; and 17) mean square root (MSR) [15].

Note that the number of channels, spectral passband during signal acquisition, sampling rates, details of preprocessing, data segmentation methods, and selected classifiers are different in previous studies. In addition, previous studies used different validation methods (within-day or cross-day). Here, we only investigated the performance differences resulting from the selected features because one of the contributions of our work is to propose skewness as a supplementary representation of sEMG biometrics. The representations of biometrics are one of the most important factors in identification performance. For a fair comparison between different sEMG features as representations, our validation methods (cross-day validation selected in our work) and all other processing steps (except feature extraction) are completely the same.

D. Validation Protocols

In our work, we designed three validation protocols to evaluate personal identification performance.

Protocol 1: In this protocol, we evaluated personal identification accuracy when all subjects employed HD-sEMG under the same gesture (or gesture combination) as the biometric token. A N-length gesture sequence was randomly generated. The 512 HD-sEMG feature vectors corresponding to each gesture were concatenated following the random gesture order to obtain a $512 \times N$-length feature vector. All 20 subjects employed the same gesture sequence. For dynamic tasks of 1-s duration, the sequence length was varied from $1 \leq N \leq 12$ (maximum total duration of 12 s). For maintenance tasks of 4-s duration, the sequence length was varied from $1 \leq N \leq 3$ (maximum total duration of 12 s). For each sequence length, 20 random gesture sequences were generated (with the replacement of each gesture) and analyzed. The average performance of all 20 repetitions was reported. In each repetition, data in sessions 1 and 2 were allocated into training and testing, respectively. Note that we had multiple recordings of each gesture per subject. To represent within-subject feature variation, tasks corresponding to the same gesture were selected both randomly and independently when generating each training example and testing example. This random selection of tasks was repeated 100 times to increase the number of training and testing examples (100 training and testing examples) for every single category to classify.

Protocol 2: In this protocol, we evaluated personal identification accuracy when each of the 20 subjects employed HD-sEMG using their own gesture (or gesture combination) as the biometric token. One random, independent gesture sequence was generated per subject (with replacement). The random generation of a gesture sequence was repeated 20 times for each subject, with average performance reported. For each generated gesture sequence (i.e., every single

category to classify), tasks corresponding to the same gesture were selected both randomly and independently. The only difference between protocols 1 and 2 is that each subject used their own unique gesture sequence to encode their HD-sEMG patterns in protocol 2. All other settings were the same as protocol 1.

Protocol 3: In this protocol, we evaluated the cancelability and cross-application discrepancy of HD-sEMG biometrics. We enrolled M accounts ($M \in \{1, 2, \ldots, 10\}$) per person using HD-sEMG with different hand gestures (or gesture combinations) as biometric tokens. The gesture sequences for each subject's account were generated randomly and independently (with replacement). In this protocol, a total of $20 \times M$ biometric tokens (20 subjects $\times$ M accounts) were registered in the personal identification system. For each account of each subject, the random gesture sequence generation, for each sequence length, was repeated 20 times, with average performance reported. For each generated gesture sequence, tasks corresponding to the same gesture were selected both randomly and independently. The only difference between protocols 2 and 3 is that each subject registered multiple accounts using different gesture sequences in protocol 3. All other settings were the same as protocol 2.

Protocols With Unknown Identities: Previous studies on sEMG biometric identification ignored the situation wherein new unknown identities aim to intrude the system. In this case, testing examples from new unknown subjects will be classified into the identity category with the highest matching score although they have not been enrolled into the system. In this work, we investigated the impact of unknown identities for the first time. An "authentication + identification" procedure was performed. We employed a fivefold cross-validation strategy. Each time, we held out four out of 20 subjects as unknown subjects, with the other 16 subjects enrolled in the system. The SVM was trained using data from the 16 subjects. A matching score was calculated before giving the identity decision. Specifically, for K enrolled accounts (one subject may enroll multiple accounts), a total of K binary SVM models were trained, with each individual account and all other accounts as the two categories to classify for each binary SVM. The kth binary SVM gives a decision hyperplane taking the following form:

$$f_k(\vec{x}) = \vec{\omega}_k^T \vec{x} + b_k = 0, \quad k \in \{1, 2, \ldots, K\} \tag{7}$$

where $\vec{x}$ is the feature vector of testing example, and $\vec{\omega}_k^T$ and b_k are model parameters. The matching score was calculated via

$$\text{score} = \max([f_1(\vec{x}), f_2(\vec{x}), \ldots, f_k(\vec{x}), \ldots, f_K(\vec{x})]). \tag{8}$$

A threshold can be selected to determine whether the testing example is similar enough to the enrolled training examples. A higher threshold will lead to false rejection of enrolled subjects. A lower threshold will lead to false acceptance of unknown subjects. The threshold at which false accept rate (FAR) equals false rejection rate (FRR) was selected. This equal error rate (EER), i.e., when FAR = FRR, was reported. If the testing example successfully passed the above authentication procedure, the identification procedure was then performed to give the identity label of the testing example. For the unknown subjects, rejection from the system refers to a correct decision. For enrolled subjects, accept into the system with the true identity label refer to a correct decision. The accuracies of decision-making in protocols 1–3 are reported. For the identification procedure, all settings were the same with protocols 1–3, except that data from 16 subjects were used to train the SVM classifiers. In this work, we aim to investigate performance degeneration when taking unknown identities into consideration. Accordingly, we first report results without unknown identities and then report results with unknown identities for comparison.

E. Relation Between the Entropy of Biometric Template and the Length of Gesture Password

The entropy of the template of a biometric modality is an essential indicator to measure the security of the biometrics against a brutal attack. So far, no previous study has investigated the relation between the entropy of the HD-sEMG biometric template and the length of gesture password, under protocol 1 (all users employed the same gesture password to encode the biometrics).

In a brutal attack, the success probability of each attack is defined as

$$\mathcal{P} = \frac{V_{\text{target}}}{V_{\text{total}}} \tag{9}$$

where V_{target} and V_{total} are the volume of feature space constructed by the target user and a large group of people, respectively. V_{total} is a hypercube (the same assumption as the previous study [19] in electroencephalogram (EEG) biometrics analysis). The shape of V_{target} is determined by different applied machine learning algorithms. To make the analysis results more general to all machine learning algorithms, we assume a hypercube shape of V_{target}. Accordingly, V_{target} and V_{total} can be obtained via calculating the side of the hypercube corresponding to each feature. We also assume that the feature spaces follow uniform distribution, and both the feature type and gesture password are known to impostors. In practical applications, the dependence between features may result in overestimation of the entropy analysis because impostors can infer the latent information of a feature subset from another feature subset. We applied the principal component analysis to remove the dependence between features. With the length of the gesture password increasing, the length of the feature vector also increases. Directly performed PCA on the long-length feature vectors is demanding. Our strategy is to first perform PCA on each 256-length feature vector corresponding to each type of the two features (VCF or skewness) and each hand gesture in the gesture password to remove the dependence between the 256 channels. We then concatenate all PCA-processed feature vectors. A second PCA was applied to the concatenated feature vector to remove dependence between different feature types and hand gestures. The entropy analysis was then performed on the dependence-removed feature space. In each PCA operation, 99% variance was preserved. The security strength of biometrics is then measured via password

entropy (same as [19]). The entropy **E** (bits) was calculated via

$$\mathbf{E} = \log_2\left(\frac{1}{\mathcal{P}}\right). \tag{10}$$

The data corresponding to each length of hand gestures in entropy analysis were the same as the counterpart used in protocol 1. The only difference was that we pool data acquired in sessions 1 and 2 together to take the cross-day variation of HD-sEMG patterns into consideration.

F. Ablation Experiment

In the ablation experiment, we assessed the necessity and contribution of each of the four components included in our method: VCF feature, skewness feature, outlier smoothing, and feature normalization. Specifically, each time, we omitted one of the four components from our method, retaining all others. The performance variation with each component omitted was used to assess the necessity and contribution of that specific component.

G. Evaluation of Computational Efficiency

To evaluate the computational efficiency of the biometric identification system (using the above-stated software and processor), the longest computation delay of all dependent processing steps was considered. If multiple processing steps are independent (e.g., feature extraction in each channel), they can be achieved via parallel computing and the longest delay considered. The dependent processing steps consisted of: 1) feature extraction; 2) detecting and smoothing corrupted channels; and 3) SVM-based decision-making. These three processing steps cannot be computed in parallel. In addition, for the feature extraction step, if multiple gestures were performed to form a gesture password, only the computation time for feature extraction of the last gesture was considered because gestures are performed sequentially (one by one) when entering the gesture password. Feature extraction of each gesture can be processed separately after that gesture is performed (in parallel with the acquisition of the next gesture). Accordingly, the ultimate computation delay is only determined by the computation on HD-sEMG of the last gesture in the sequence. The computation time $\mathtt{T}$ with the increasing length of gesture password $\mathtt{L}$ can be described as

$$\mathtt{T}(\mathtt{L}) = \mathtt{T}_{\text{dynamic}}(\mathtt{L}) + \mathtt{T}_{\text{inherent}} + \mathtt{T}_{\text{random}} \tag{11}$$

where $T_{\text{dynamic}}(\mathtt{L})$ denotes the computation time-varying with different lengths of gesture passwords (e.g., the computation time of SVM-based decision-making with different lengths of feature vectors constructed by multiple gestures), $\mathtt{T}_{\text{inherent}}$ denotes the computation time of inherent processing steps that are independent of $\mathtt{L}$ (e.g., parallel computing of feature extraction for each gesture), and $\mathtt{T}_{\text{random}}$ denotes the computation time resulting from random factors (e.g., time-varying conditions of the processor). The computational time of each protocol was evaluated separately. The average results of 100 repetitions are reported.

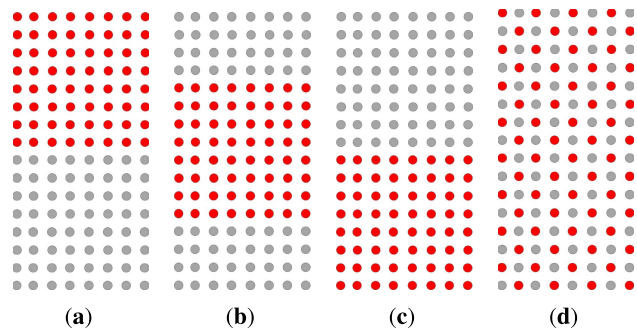


Fig. 5. Different electrode layouts with reduced number of electrodes. Red represents the selected active electrodes, whereas gray indicates unused electrodes. In all presented layouts, a total of 128 electrodes are used. (a) Layout 1: distal. (b) Layout 2: middle. (c) Layout 3: proximal. (d) Layout 4: sparse.

H. Comparison Between Different Electrode Layouts

Our work used forearm HD-sEMG as biometrics with a large number of HD electrodes. We aim to compare the effects of: 1) a large area covered by sparsely distributed electrodes and 2) densely distributed electrodes within a small area. Such comparison can help to find the most crucial factor and, thereby, better explain the obtained high identification accuracy from the aspect of data sensing.

In addition to evaluating the identification performance using all 256 channels, we also investigated the performance variation with four different electrode layouts, as presented in Fig. 5. The first three layouts measure HD-sEMG from the distal, middle, and proximal parts of a subject's forearm, respectively, with densely distributed electrodes. The last electrode layout measures HD-sEMG from the whole forearm with sparsely distributed electrodes. The performance of all these electrode layouts will be compared.

I. Statistical Analysis

The statistical analysis was applied to quantify the difference in achieved results due to a specific factor. Because the obtained results do not follow a Gaussian distribution, the two-sided sign test [20], a nonparametric method, was employed. To avoid multiple comparison errors, the Bonferroni–Holm correction [21] was also applied. If not specified, only p-values with the Bonferroni–Holm correction are reported (when correction is required). A significant difference was claimed if $p < 0.05$ was obtained.

IV. Results

A. Identification Accuracy Versus Gesture Sequence Length

Fig. 6 presents average identification accuracy versus gesture sequence length in each protocol. No significant differences were found between the accuracies of dynamic and maintenance tasks under the same validation protocol (if not specified, $M = 2$ was analyzed in protocol 3) with the same number of tasks. Maintenance tasks require an initial period to reach the target gesture, increasing the total duration to perform the gesture sequence in practical use. A longer duration to perform the gesture sequence may reduce convenience

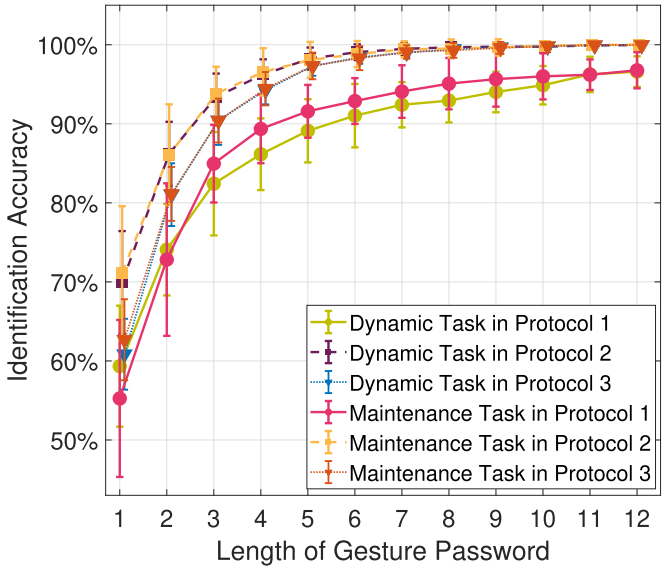


Fig. 6. Average identification accuracy versus gesture sequence length in each protocol ($M = 2$ in protocol 3). Considering that maintenance tasks had a longer duration (4 s), for a fair comparison, only signals from 1.25 to 2 s (0.75-s duration) during maintenance tasks were used in our analysis. For symbol plotting of protocols 2 and 3, a slight shift with horizontal shift values of 0.05 and 0.1, respectively, was added to avoid overlap of sample points.

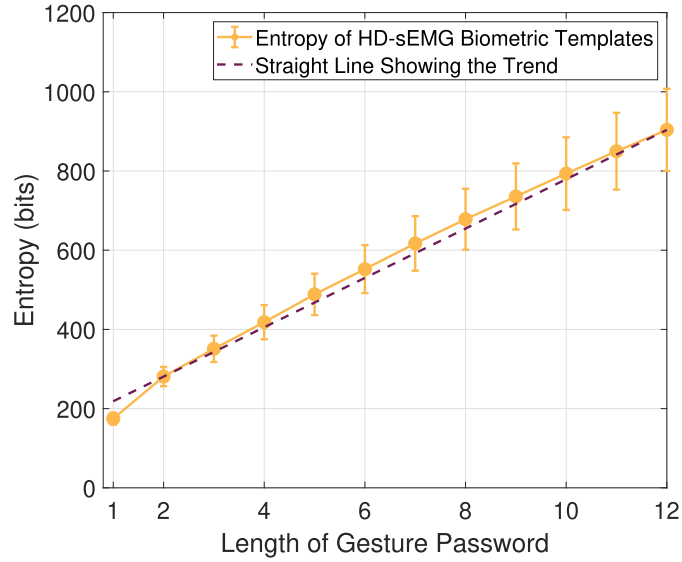


Fig. 7. Entropy of HD-sEMG biometric template with varying length of gesture password.

in real-world applications. Accordingly, only the results of dynamic tasks are shown hereafter.

According to Fig. 6, identification accuracy increases with the gesture sequence length. Results of statistical analysis also support this claim. The identification accuracy using a 12-length dynamic task sequence is significantly higher than the counterpart using a 1-length dynamic task sequence, with $p < 0.01$ achieved for all protocols 1–3. With a 1-length gesture sequence using dynamic tasks, the proposed method achieved 59.33% ± 7.65%, 69.93% ± 6.50%, and 60.85% ± 4.48% identification accuracy for protocols 1–3, respectively. With a 12-length gesture sequence using dynamic tasks, the proposed method achieved 96.60% ± 1.93%, 99.96% ± 0.12%, and 99.96% ± 0.05% identification accuracies for protocols 1–3, respectively. Protocol 1 is the most challenging, with the lowest identification accuracy achieved ($p < 0.05$ achieved for both comparison between protocols 1 and 2 and comparison between protocols 1 and 3 with a 12-length gesture sequence).

B. Results of Entropy Analysis

According to Fig. 7, using the proposed feature set, HD-sEMG biometric template encoded by 1-length gesture sequence is with 174.7 ± 15.1 bits entropy. Assuming that each attempt of brutal attack takes 1000 flops, using a 10-petaFLOPS supercomputer to break our biometric identification system takes $((2^{174.7} \times 10^3)/(10 \times 10^{15})) \approx 3.89 \times 10^{39}$ s, i.e., $\approx 1.23 \times 10^{32}$ years, which can be viewed as impossible with current advancement of computation. This evaluation is similar to a previous study [19] on the security analysis of EEG biometrics. Moreover, with the length of the gesture password increasing, the entropy shows an approximately linear growth. With the length of the gesture password increasing by 1, the entropy increases by 66.29 ± 14.35 bits on average. These results show that the proposed HD-sEMG biometrics is chaotic enough and highly secure against a brutal attack. The approximately linear growth of entropy with the increase in the length of gesture sequence demonstrates the independence of information provided by different gestures, which, in turn, proves the effectiveness of biometric encoding via gesture password (even if all users enter the same gesture password).

C. Results of Ablation Experiment

We performed the ablation experiment under all protocols ($M = 2$ in protocol 3). Mean ± standard deviation results of the ablation experiment using 12-length gesture sequences of dynamic tasks are shown in Table I. With one specific component omitted, the average identification accuracy significantly decreased in most cases. Moreover, omitting feature normalization has the greatest (adverse) impact on identification accuracy. Although feature normalization is a common choice in many applications, these results demonstrate the necessity of feature normalization in the emerging area of sEMG biometric identification.

D. Identification Accuracy Versus Increasing Number of Accounts

Protocol 3 tested the performance of the proposed method when each subject enrolled multiple accounts in different applications. With the length of gesture password fixed at 3 and, separately, 12, we further investigated the accuracy variation due to changing the number of accounts for each subject (with the number of subjects fixed at 20) and changing the number of subjects (with the number of accounts for each subject fixed at 10). Results are presented in Fig. 8. Significant accuracy differences were found between 200- and 40-account identification tasks in all cases with the higher

TABLE I

MEAN ± STANDARD DEVIATION ACCURACY (%) VARIATION IN ABLATION EXPERIMENT WITH A 12-LENGTH SEQUENCE OF DYNAMIC TASKS. $M = 2$ IN PROTOCOL 3. SYMBOL ↓ DENOTES A REDUCED IDENTIFICATION ACCURACY WITH A SPECIFIC COMPONENT OMITTED. SYMBOL * REFERS TO A SIGNIFICANT DIFFERENCE COMPARED WITH THE COUNTERPARTS USING ALL COMPONENTS

Protocol	Settings	Length=2	Length=4	Length=6	Length=8	Length=10	Length=12
Protocol 1	Keep All Components	74.1±5.8	86.1±4.5	91.0±4.0	92.9±2.8	94.9±2.4	96.6±1.9
	Omit Skewness Feature	71.7±5.3↓*	84.2±4.4↓*	88.7±4.5↓*	91.1±3.6↓	93.6±2.5↓	94.7±2.8↓
	Omit VCF Feature	44.0±5.9↓*	56.4±5.3↓*	65.2±4.8↓*	70.6±3.6↓*	73.9±3.9↓*	76.7±4.5↓*
	Omit Outlier Smoothing	51.3±8.0↓*	59.2±4.7↓*	64.0±3.9↓*	65.9±3.6↓*	67.7±3.4↓*	68.9±3.9↓*
	Omit Feature Normalization	7.1±2.7↓*	11.5±7.5↓*	22.1±13.5↓*	33.0±13.8↓*	35.3±16.1↓*	43.0±17.0↓*
Protocol 2	Keep All Components	86.2±4.0	96.2±1.9	99.1±1.0	99.7±0.5	99.8±0.5	99.96±0.12
	Omit Skewness Feature	83.6±5.8↓	94.3±2.8↓	98.1±1.6↓*	99.3±1.0↓	99.5±0.7↓	99.8±0.3↓
	Omit VCF Feature	53.7±6.8↓*	72.8±5.0↓*	81.3±4.5↓*	87.5±3.1↓*	90.8±2.6↓*	92.8±2.6↓*
	Omit Outlier Smoothing	66.2±4.6↓*	76.6±5.0↓*	81.0±4.5↓*	84.2±4.2↓*	86.0±3.7↓*	88.1±3.4↓*
	Omit Feature Normalization	8.3±4.2↓*	25.4±18.5↓*	31.9±21.5↓*	53.7±20.7↓*	49.2±22.9↓*	59.5±21.1↓*
Protocol 3	Keep All Components	81.0±4.0	94.3±1.8	98.4±0.8	99.5±0.4	99.8±0.3	99.96±0.05
	Omit Skewness Feature	78.7±4.4↓*	92.1±2.3↓*	97.2±1.3↓*	98.8±0.6↓*	99.4±0.5↓*	99.8±0.2↓*
	Omit VCF Feature	45.4±4.9↓*	65.4±3.7↓*	76.7±3.2↓*	83.4±2.8↓*	87.8±1.9↓*	90.5±1.5↓*
	Omit Outlier Smoothing	64.2±3.4↓*	78.8±3.4↓*	85.7±2.5↓*	89.7±1.8↓*	91.2±2.2↓*	92.6±1.5↓*
	Omit Feature Normalization	2.6±1.7↓*	14.0±11.9↓*	27.1±16.4↓*	35.2±19.5↓*	42.0±21.0↓*	58.5±15.7↓*

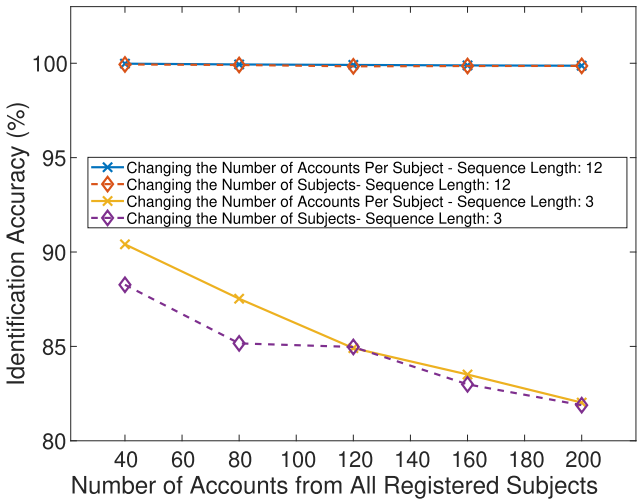


Fig. 8. Identification accuracy variation with the increasing account number (changing the number of subjects or the number of accounts for each subject). When changing the number of subjects (from four subjects to 20 subjects with an increment of four subjects), ten accounts were registered for each subject. When changing the number of accounts for each subject (from two accounts to ten accounts with an increment of two accounts), all 20 subjects were registered.

number of accounts providing less accuracy ($p = 0.012$ for the case of changing the number of subjects with a 12-length gesture sequence and $p < 0.001$ for all other three cases). However, for the 12-length gesture sequence, the accuracy showed relatively slow decreasing trends as the number of all accounts increased, with the accuracy maintaining a relatively high level of >99.8%.

E. Performance Degeneration With Unknown Identities

The results using four of 20 unknown subjects are presented in Table II. With unknown identities, the identification accuracy decreased from 96.60% to 81.44% (EER: 0.1818) and from 99.96% to 95.37% (EER: 0.0459) in protocols 1 and 2, respectively. With unknown identities, the identification accuracy in protocol 3 decreased from 99.96% to 93.82% (EER: 0.0615) for 40-account classification [(16 enrolled subjects + 4 unknown subjects) × 2 accounts per subject] and decreased from 99.85% to 93.81% (EER: 0.0611) for 200-account classification [(16 enrolled subjects + 4 unknown subjects) × 10 accounts per subject]. In all cases, accuracy decreased significantly when unknown identities were inserted ($p < 0.01$).

TABLE II

PERFORMANCE VARIATION WITH UNKNOWN IDENTITIES. ACCURACY (ACC), SENSITIVITY (SN), AND SPECIFICITY (SP) WERE REPORTED. FOR THE CASES WITH UNKNOWN IDENTITIES, EER VALUES WERE ALSO REPORTED, AND THE UNKNOWN IDENTITY WAS ASSIGNED AN ADDITIONAL LABEL TO EVALUATE THE SENSITIVITY AND SPECIFICITY. SENSITIVITY AND SPECIFICITY VALUES WERE WEIGHTED AVERAGE OF EACH CLASS (ACCORDING TO THE NUMBER OF EXAMPLES OF EACH CLASS). ALL RESULTS WERE OBTAINED WITH 12-LENGTH GESTURE PASSWORD

Protocol	Without Unknown Identities (20 Enrolled Subjects)	With Unknown Identities (16 Enrolled Subjects + 4 Unknown Subjects)
Protocol 1	Acc: 96.60% Sn: 96.60% Sp: 99.82%	Acc: 81.44% Sn: 81.27% Sp: 96.10% EER: 18.18%
Protocol 2	Acc: 99.96% Sn: 99.95% Sp: 100%	Acc: 95.37% Sn: 95.51% Sp:99.05% EER: 4.59%
Protocol 3 (2 Accounts per Subject)	Acc: 99.96% Sn: 99.97% Sp: 100%	Acc: 93.82% Sn: 93.81% Sp: 98.73% EER: 6.15%
Protocol 3 (10 Accounts per Subject)	Acc: 99.85% Sn: 99.85% Sp: 100%	Acc: 93.81% Sn: 93.80% Sp: 98.74% EER: 6.11%

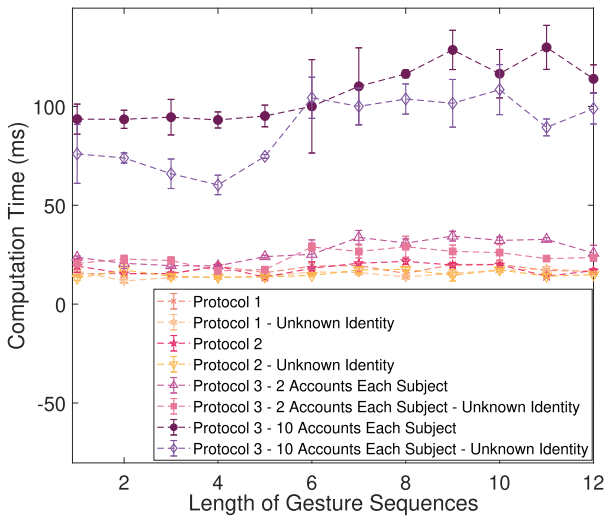


Fig. 9. Computation delay in each protocol.

F. Results of Computation Efficiency

Fig. 9 presents the computation time of the proposed system under each protocol. The computation time does not increase linearly with gesture password length, probably due to the variation of terms $\mathrm{T}_{\mathrm{inherent}}$ and $\mathrm{T}_{\mathrm{random}}$ in (11). However, overall, longer gesture passwords led to a general trend of increased computation time. We calculated the average computation time of 1–6-length gesture passwords and 7–12-length gesture passwords, under protocol 3 (ten accounts each subject without an unknown identity). Statistical analysis showed that longer gesture passwords resulted in a significantly longer computation time compared with the shorter ones ($p < 0.01$). However, even in the worst case, the computational time is less than 130 ms, which can support real-time identification in practical use.

G. Results of Comparison Between the Features in Our Study and Previous Studies

Table III presents the identification accuracy using each single feature. The VCF feature, which was proposed to be used as sEMG biometric representations for the first time in our previous work [13], achieved the best performance with significantly better performance compared with the second best feature (SSC).

Considering that investigating all possible feature combinations is computationally intractable, we fixed the optimal feature (VCF) and added each one of the other features as supplementary sEMG biometrics representations. Table IV presents the identification results using the combination of two features. Although skewness did not produce excellent performance individually, when combined with VCF, it is the best supplementary feature. The combination of VCF and skewness produced the highest identification accuracy of 96.60%. When comparing the accuracy using VCF and skewness with the counterparts using all other feature combinations listed in Table IV, significant differences were found in all cases except the combination of VCF and WL ($p = 0.1435$). We further fixed the combination of VCF and skewness and added a third feature. In all cases, the obtained accuracy was lower than 96.60% with the highest accuracy of 96.52% obtained by the combination of VCF, skewness, and SampEn features (no significant difference was found compared with the combination of VCF and skewness). These results demonstrate that the combination of VCF and skewness is a powerful representation of sEMG biometrics.

TABLE III
PERFORMANCE OF EACH SINGLE FEATURE (USING 12-LENGTH GESTURE PASSWORD UNDER PROTOCOL 1)

No.	Feature	Accuracy
1	VCF	94.68%
2	SSC	90.87%
3	Hjorth2	90.16%
4	SpEn	89,79%
5	MDF	87.22%
6	ZC	85.90%
7	SampEn	84.25%
8	WL	81.91%
9	DASDV	78.58%
10	Skewness	76.75%
11	LOGD	76.56%
12	MAV	75.62%
13	RMS	75.56%
14	SSI	72.48%
15	Hjorth1	72.37%
16	TM4	65.00%
17	MSR	62.65%
18	TM3	59.21%
19	TM5	57.96%

TABLE IV
PERFORMANCE OF EACH FEATURE COMBINED WITH THE OPTIMAL VCF FEATURE (USING 12-LENGTH GESTURE PASSWORD UNDER PROTOCOL 1)

No.	Feature	Accuracy
1	VCF+Skewness	96.60%
2	VCF+WL	94.64%
3	VCF+SpEn	94.50%
4	VCF+MDF	94.49%
5	VCF+Hjorth2	94.30%
6	VCF+DASDV	94.12%
7	VCF+SampEn	93.88%
8	VCF+SSC	93.49%
9	VCF+ZC	92.17%
10	VCF+LOGD	91.74%
11	VCF+MAV	90.84%
12	VCF+RMS	88.93%
13	VCF+SSI	85.87%
14	VCF+Hjorth1	85.68%
15	VCF+TM3	83.91%
16	VCF+MSR	83.19%
17	VCF+TM5	82.92%
18	VCF+TM4	81.10%

H. Results of Comparison Between Different Electrode Layouts

Fig. 10 presents the accuracy results for the different electrode layouts and lengths of gesture sequences. Layouts 1–3, with a small area covered by densely distributed electrodes, led to a significantly lower identification accuracy compared with layouts 4 and 5 with a large area covered by either sparsely (layout 4) or densely (layout 5) distributed electrodes for all lengths of gesture sequences from 1 to 12. Reducing

TABLE V
REVIEW OF CURRENT sEMG BIOMETRICS STUDIES

Study	Year of Publication	Number of Subjects	Limitations
[22]	2015	14 subjects	Training and testing data were acquired on the same day. The factor of unknown identities was not investigated in biometric identification.
[15]	2020	10 subjects	Only authentication tasks (binary decision making) were considered. Challenging identification tasks with unknown identities were not considered. Training and testing data were acquired on the same day. The number of subjects is relatively low.
[23]	2020	24 subjects	Training and testing data were acquired on the same day. The factor of unknown identities was not investigated in biometric identification.
[17]	2020	22 subjects	Training and testing data were acquired on the same day. The factor of unknown identities was not investigated in biometric identification.
[7]	2020	22 subjects	Only authentication tasks (binary decision making) were considered. Challenging identification tasks with unknown identities were not considered. Training and testing data were acquired on the same day. Using EMG under isometric contractions (unobservable thereby secure, but unnatural and uncomfortable) as biometrics.
[10]	2020	22 subjects	The factor of unknown identities was not investigated in biometric identification. Using EMG under isometric contractions (unobservable thereby secure, but unnatural and uncomfortable) as biometrics. The cross-day identification accuracy (85.8%) was relatively low.
[11]	2021	24 subjects	Training and testing data were acquired on the same day. The factor of unknown identities was not investigated in biometric identification.
[12]	2021	24 subjects	Training and testing data were acquired on the same day. The factor of unknown identities was not investigated in biometric identification.
[13]	2021	20 subjects	Only authentication tasks (binary classification) were considered.
[24]	2021	22 subjects	Using EMG under isometric contractions (unobservable thereby secure, but unnatural and uncomfortable) as biometrics. The cross-day identification accuracy (87.3%) was relatively low.
This work	-	20 subjects	EMG biometrics acquired from a total of 256 electrodes require more computational resources and higher power consumption.

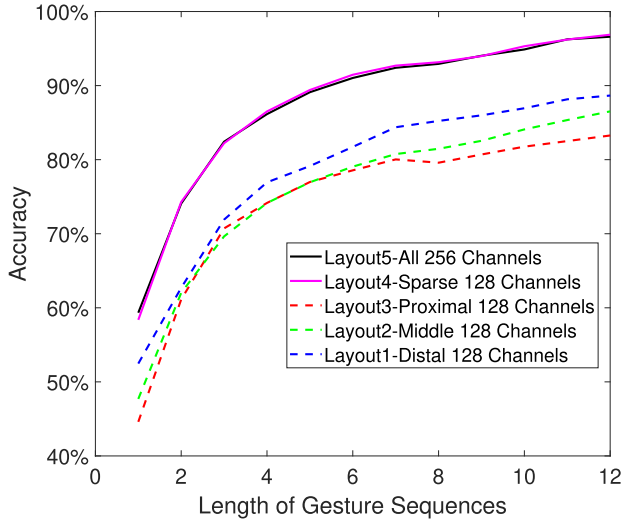


Fig. 10. Identification accuracy of each electrode layout as a function of the gesture length.

the density of electrodes (layout 4) to half of the original level (layout 5) while keeping the total range of forearm covered by electrode arrays did not result in significantly different identification accuracy.

V. DISCUSSION

A. Comparison With Current State-of-the-Art EMG Biometrics-Based Personal Identification

EMG biometrics is an emerging approach for cancelable and cross-application discrepant personal identification. A review of current sEMG biometrics studies is presented in Table V. Validation methods in previous studies on EMG biometrics can be divided into two categories, namely, single-day validation [11], [12], [17], [22], [23] and cross-day validation [10]. For the single-day validation, training and testing data were acquired on the same day. Accordingly, the variation of signal characteristics was not considered. Moreover, the training and testing data may be acquired in the same session without the replacement of electrodes. In this case, the electrode shift was also ignored, which is a key factor in EMG pattern recognition [25]. Therefore, the single-day validation may largely overestimate the performance in real-world applications. For the cross-day validation, training and testing data were acquired on different days. Signal variation across days and electrode shift across signal acquisition sessions were considered, thus providing a more reliable performance evaluation for practical use. Considering our work employed the cross-day validation strategy, we compared our results with previous studies with the same validation strategy. Because EMG biometrics-based personal identification is an emerging application, only limited previous studies employed the cross-day validation. The previous study [10] yielded much lower identification accuracies <88% on the identification of enrolled 4–44 accounts, all using 24-s signal duration and 64 channels. By contrast, this work achieved an identification accuracy of 99.85% in recognizing 200 accounts. Even with unknown identities, the present work achieved an identification accuracy of 93.81% for 200-account tasks [(16 enrolled subjects + 4 unknown subjects) × 10 accounts per subject], still higher than the previous study.

Possible explanations for the better performance in this work compared with our previous studies are considered from five aspects.

1) From the aspect of muscle anatomy, a subject's forearm contains more functioning muscle groups during most hand and wrist activities of daily life. The dorsal hand location used in our previous study [10] is limited to intrinsic hand muscles only. Accordingly, the acquired forearm EMG signals in this work were more informative. In addition, the extensor digitorum muscle of the forearm is cylinder-shaped, located within the forearm along the proximal-distal direction. Different muscle compartments have oblique fascicles overlapping with each other, rather than running in parallel [26]. This property contributes to a highly complex and highly intersubject different anatomical structure. Different anatomical structures (e.g., different volume conductor properties determined by the thickness of subcutaneous tissues) further result in the variation of sEMG characteristics for different subjects.
2) From the aspect of human gesture kinematics, dynamic contraction, compared with isometric contraction, is a more natural and comfortable contraction style. All subjects repeatedly perform our selected hand gestures by natural dynamic muscle contractions, so the patterns of these gestures should be relatively fixed for each specific subject but varied between subjects, further contributing to the intersubject differences in sEMG characteristics.
3) From the aspect of hardware sensing, we acquired 256-channel HD-sEMG in this work covering almost the whole forearm to capture the activation map of forearm muscles. Also, the larger area covered by HD-sEMG electrode arrays likely reduced the negative impact of a slight electrode shift when replaced on a second day, reducing the cross-day variation of sEMG patterns. We also provided evidence on the improvement of system performance using electrodes covering a large forearm area by comparing different electrode layouts presented in Fig. 10. In practical use with fixed computational and data acquisition resources (i.e., a fixed number of electrodes), we could distribute all electrodes across a large area. These findings can be explained by the relatively independent and, thereby, supplementary information in channels distributed in different locations over a wide range of muscle groups. Generally, an outlier feature in a corrupted electrode would have a higher impact on system performance in systems with a lower number of total electrodes. However, with a larger number of electrodes, the chance that an electrode is corrupted is also much higher. The mathematical expectation of the proportion of outlier features in corrupted electrodes would not change with a different number of electrodes. Accordingly, a higher number of electrodes is not a crucial factor compared with a larger measuring area. With a large measuring area, the identification accuracy can reach an extremely high level ($>99\%$). With many more unstudied muscle groups in the whole human body, the potential of sEMG biometrics is extremely high. A significantly improved identification accuracy can be expected with more convenient textile-based wearable sEMG acquisition integrated into clothing [27] to acquire sEMG from more body areas.
4) From the aspect of experimental design, we studied 34 hand gestures in this work, forming a large pool of alternative sEMG biometric tokens encoded by different gestures. Selecting multiple and different gestures to encode HD-sEMG biometrics further increases the unique pattern of HD-sEMG biometrics.
5) Moreover, from the technical aspect, the combination of VCF and skewness features used as sEMG biometric representations showed superiority in comparison to results with other features. VCF is the most effective single feature. The skewness feature, first proposed as sEMG biometric representations in this work, is a simple but powerful supplementary feature when combined with VCF. Directly using skewness, a statistical parameter of sEMG signals, as biometric representations can achieve good performance, indicating the high potential of sEMG as biometrics. More powerful sEMG biometric representations can be investigated in future studies. The above five key points show the superiority of this study compared with previous studies. Furthermore, this work properly integrated personal authentication and identification tasks together by taking the factor of unknown identities (unregistered persons) into consideration. By reliably rejecting unregistered persons, this system is more suitable for realistic applications compared with the previous one [10].

Our work demonstrated that spreading the available electrode over a larger range of muscle groups can significantly improve identification performance. Although the 256 electrodes used in this work required more computational resources and higher power consumption during signal acquisition, this limitation can hopefully be addressed by advanced wearable and compressed sEMG sensing [27], [28] and low-power computation. Currently, demonstrating the potential of an extremely high sEMG biometric identification accuracy is the key step to promote an increase in future studies in this area.

So far, no consensus exists on the optimal processing pipeline for sEMG biometric identification, partly due to the fact that sEMG biometric identification is an emerging area, and partly due to the lack of open-access cross-day sEMG biometrics datasets available for a fair comparison between the yielded results in different studies. Our dataset, which consists of data from 20 subjects with each subject participating in our experiment on two separate days, has been open-sourced previously [18]. We hope that our dataset can further contribute to more future sEMG biometrics studies and provide a unique platform for comparison between different methods in a rigorous cross-day validation. In addition, we compared different features used in our work and previous studies, providing references on identification performance from the aspect of sEMG representations. We proved that, with sEMG signals acquired from all muscle groups from the forearm, sEMG

biometrics can achieve an excellent cross-day identification performance.

B. Component Employed in Our Proposed Method

In our method, four components were employed to achieve a high identification accuracy. The ablation experiment indicates the necessity of all employed components. Skewness and VCF features represent the HD-sEMG biometric template in the temporal and spectral domains, respectively. All features were extracted from each of the 256 channels, further representing the HD-sEMG biometric template in the spatial domain. These temporal–spectral–spatial-domain features contain sufficient information on the individually unique muscle activation patterns. The outlier smoothing component can reduce the impact of outliers extracted from those bad channels not active during data acquisition. Outliers in those channels not working well may largely degenerate identification performance [29]. With outliers smoothed by their neighbor channels, the proposed method can achieve more robust performance, with the accuracy remarkably improved in all protocols. The feature normalization component can balance the variance of each single constituent feature vector. The constituent feature vectors with a higher standard deviation may dominate when calculating the distance between two examples in the feature space. Equivalently, the number of effective features was reduced. With each constituent feature vector normalized to the same standard deviation, all constituent feature vectors make equal contributions to the distance in the feature space between two examples, thus improving the identification accuracy.

C. Superiority of the HD-sEMG Biometric Modality

As aforementioned, the HD-sEMG biometric modality is cancelable, thus supporting cross-application discrepant personal identification. This unique property can make up for the flaws of most existing biometrics. The HD-sEMG biometric template is also theft-resistant. Compared with face, DNA, gait, and fingerprint identification, an HD-sEMG template is almost impossible to steal without the knowledge of users because the HD-sEMG electrode array must be in close contact with a user's skin to acquire valid HD-sEMG signals. In addition, HD-sEMG biometrics can protect users' volition. Specifically, users must perform muscle contractions volitionally to generate sEMG signals because sEMG is the summation of motor unit spike trains originating from spinal nerves. By contrast, users can be forced to enter their fingerprint (or other biometrics, such as face and iris) because the compulsive fingerprint patterns show no differences compared with those entered voluntarily. Moreover, HD-sEMG biometrics can be further encoded by performing different muscle contraction tasks (i.e., different hand gestures). For a 200-account recognition task, the proposed method achieved a high accuracy of 99.85% and 93.81% without and with unknown identities, respectively, indicating the promising prospect of HD-sEMG biometrics. While a multiple gesture password, as described herein, might be time-consuming, shorter gesture durations and elimination/shortening of rest transitions could greatly reduce this time. The time required to don and doff an HD-sEMG system is also currently too large for most biometrics applications. However, embedded wearable HD-sEMG systems are currently under development [30], which might help resolve this concern in the very near future. By involving more gestures and muscles from other body parts and/or including other biometrics concurrently (e.g., gait, fingerprint), identification accuracy might be further improved (and authentication duration decreased).

D. Limitations and Future Work

One important direction in future studies is to validate the performance of sEMG biometrics on a larger dataset, which is an essential factor to be investigated before real-world applications. In addition, although we investigated the effects of different electrode layouts and demonstrated electrode layout as an important factor in identification accuracy, we did not present the optimal layout of electrodes. Future efforts should be put into finding the optimal electrode layout (e.g., based on the anatomy of muscles), so as to reduce the required number of electrodes needed to achieve the desired identification performance.

VI. Conclusion

In our work, we proposed cancelable HD-sEMG-based biometrics for cross-application discrepant personal identification. We achieved a high accuracy of 99.96% and 99.85% for the 40-account (20 subjects $\times$ 2 accounts) and 200-account (20 subjects $\times$ 10 accounts) recognition tasks, respectively, without unknown identities. Even with unknown identities, high accuracy of 93.82% and 93.81% was achieved in the above two cases, respectively, remaining higher than current state-of-the-art performance. The proposed method can make up for the limitations of existing biometric modalities in certain application scenarios. Our work can also improve information security in BSN-based smart healthcare applications, allowing users to cancel the original biometric template once it is exposed to impostors and enrolling multiple accounts to shift between modes of healthcare services. The high accuracy of HD-sEMG biometrics can also push the limit of current advancements in the information security field, as applied to more general daily applications.

References

[1] G. Zheng *et al.*, "Finger-to-heart (F2H): Authentication for wireless implantable medical devices," *IEEE J. Biomed. Health Inform.*, vol. 23, no. 4, pp. 1546–1557, Jul. 2019.

[2] M. Ibrahim, T.-C. Liang, K. Scott, K. Chakrabarty, and R. Karri, "Molecular barcoding as a defense against benchtop biochemical attacks on DNA fingerprinting and information forensics," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 3595–3609, 2020.

[3] M. Luo, J. Cao, X. Ma, X. Zhang, and R. He, "FA-GAN: Face augmentation GAN for deformation-invariant face recognition," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 2341–2355, 2021.

[4] C. Peng, N. Wang, J. Li, and X. Gao, "Soft semantic representation for cross-domain face recognition," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 346–360, 2021.

[5] H. Sun *et al.*, "Synchronous fingerprint acquisition system based on total internal reflection and optical coherence tomography," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 10, pp. 8452–8465, Oct. 2020.

[6] S. Chen, Z. Guo, J. Feng, and J. Zhou, "An improved contact-based high-resolution palmprint image acquisition system," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 9, pp. 6816–6827, Sep. 2020.

[7] X. Jiang *et al.*, "Neuromuscular password-based user authentication," *IEEE Trans. Ind. Informat.*, vol. 17, no. 4, pp. 2641–2652, Apr. 2021.

[8] S. Zhao *et al.*, "Wearable physiological monitoring system based on electrocardiography and electromyography for upper limb rehabilitation training," *Sensors*, vol. 20, no. 17, p. 4861, Aug. 2020.

[9] Z. Lu, X. Chen, Q. Li, X. Zhang, and P. Zhou, "A hand gesture recognition framework and wearable gesture-based interaction prototype for mobile devices," *IEEE Trans. Human-Mach. Syst.*, vol. 44, no. 2, pp. 293–299, Apr. 2014.

[10] X. Jiang *et al.*, "Cancelable HD-sEMG-based biometrics for cross-application discrepant personal identification," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 4, pp. 1070–1079, Apr. 2021.

[11] A. Pradhan, J. He, and N. Jiang, "Performance optimization of surface electromyography based biometric sensing system for both verification and identification," *IEEE Sensors J.*, vol. 21, no. 19, pp. 21718–21729, Oct. 2021.

[12] A. Pradhan, J. He, and N. Jiang, "Score, rank, and decision-level fusion strategies of multicode electromyogram-based verification and identification biometrics," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 3, pp. 1068–1079, Mar. 2022.

[13] X. Jiang *et al.*, "Enhancing IoT security via cancelable HD-sEMG-based biometric authentication password, encoded by gesture," *IEEE Internet Things J.*, vol. 8, no. 22, pp. 16535–16547, Nov. 2021.

[14] R. N. Khushaba and K. Nazarpour, "Decoding HD-EMG signals for myoelectric control–how small can the analysis window size be?" *IEEE Robot. Autom. Lett.*, vol. 6, no. 4, pp. 8569–8574, Oct. 2021.

[15] Q. Li, P. Dong, and J. Zheng, "Enhancing the security of pattern unlock with surface EMG-based biometrics," *Appl. Sci.*, vol. 10, no. 2, p. 541, Jan. 2020.

[16] S. Shin, M. Kang, J. Jung, and Y. T. Kim, "Development of miniaturized wearable wristband type surface EMG measurement system for biometric authentication," *Electronics*, vol. 10, no. 8, p. 923, Apr. 2021.

[17] X. Jiang, K. Xu, X. Liu, D. Liu, C. Dai, and W. Chen, "High-density surface electromyogram-based biometrics for personal identification," in *Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2020, pp. 728–731.

[18] X. Jiang *et al.*, "Open access dataset, toolbox and benchmark processing results of high-density surface electromyogram recordings," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 1035–1046, 2021.

[19] K. Sadeghi, A. Banerjee, J. Sohankar, and S. K. S. Gupta, "Geometrical analysis of machine learning security in biometric authentication systems," in *Proc. 16th IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2017, pp. 309–314.

[20] W. J. Dixon, "Power functions of the sign test and power efficiency for normal alternatives," *Ann. Math. Statist.*, vol. 24, no. 3, pp. 467–473, Sep. 1953.

[21] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandin. J. Statist.*, vol. 6, no. 2, pp. 65–70, Jan. 1979.

[22] S. Venugopalan, F. Juefei-Xu, B. Cowley, and M. Savvides, "Electromyograph and keystroke dynamics for spoof-resistant biometric authentication," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Boston, MA, USA, Jun. 2015, pp. 109–118.

[23] J. He and N. Jiang, "Biometric from surface electromyogram (sEMG): Feasibility of user verification and identification based on gesture recognition," *Frontiers Bioeng. Biotechnol.*, vol. 8, p. 58, Feb. 2020.

[24] J. Fan *et al.*, "Cancelable HD-SEMG biometric identification via deep feature learning," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 4, pp. 1782–1793, Apr. 2022.

[25] A. Stango, F. Negro, and D. Farina, "Spatial correlation of high density EMG signals provides features robust to electrode number and shift in pattern recognition for myocontrol," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 23, no. 2, pp. 189–198, Mar. 2015.

[26] J. N. A. L. Leijnse, S. Carter, A. Gupta, and S. McCabe, "Anatomic basis for individuated surface EMG and homogeneous electrostimulation with neuroprostheses of the extensor digitorum communis," *J. Neurophysiology*, vol. 100, no. 1, pp. 64–75, Jul. 2008.

[27] A. Shafti, R. B. R. Manero, A. M. Borg, K. Althoefer, and M. J. Howard, "Designing embroidered electrodes for wearable surface electromyography," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, May 2016, pp. 172–177.

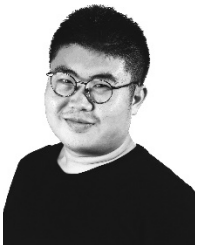
[28] M. Balouchestani and S. Krishnan, "Effective low-power wearable wireless surface EMG sensor design based on analog-compressed sensing," *Sensors*, vol. 14, no. 12, pp. 24305–24328, Dec. 2014.

[29] P. Ghaderi and H. R. Marateb, "Muscle activity map reconstruction from high density surface EMG signals with missing channels using image inpainting and surface reconstruction methods," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 7, pp. 1513–1523, Jul. 2017.

[30] G. L. Cerone, A. Botter, and M. Gazzoni, "A modular, smart, and wearable system for high density sEMG detection," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 12, pp. 3371–3380, Dec. 2019.

Xinyu Jiang received the B.Eng. degree (Hons.) in electronic engineering and the M.S. degree in circuits and systems from Fudan University, Shanghai, China, in 2018 and 2021, respectively.

His research interests include medical artificial intelligence, physiological computing, human–machine interface, and biomedical signal processing with applications to various industrial and clinical scenarios.



Xiangyu Liu (Member, IEEE) received the Ph.D. degree in industrial design from the East China University of Science and Technology, Shanghai, China, in 2021.

He is currently a Lecturer with the College of Communication and Art Design, University of Shanghai for Science and Technology, Shanghai. His research interests include neuromuscular prosthesis and exoskeleton design.

Jiahao Fan received the B.S. degree in communication engineering and the M.S. degree in electronic and communication engineering from Jilin University, Changchun, China, in 2013 and 2016, respectively, and the Ph.D. degree in biomedical engineering from the School of Information Science and Technology, Fudan University, Shanghai, China, in 2021.

His research interests are mainly on sleep pattern recognition algorithms, which mainly are automatic stage scoring and sleep disease detection. He is also interested in signal processing, machine learning, and artificial neural networks.

Xinming Ye received the bachelor's degree in physical education from East China Normal University, Shanghai, China, in June 1997, and the master's degree with a major in sports humanities and sociology from the School of Sports Science and Engineering, East China University of Science and Technology, Shanghai, in March 2007. He is currently pursuing the Ph.D. degree in law with Tongji University, Shanghai.

He is currently an Associate Professor with the School of Sports Science and Engineering, East China University of Science and Technology. His research interests include exercise and health, and nutrition and health.

Chenyun Dai (Member, IEEE) received the B.S. degree from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2010, and the M.S. and Ph.D. degrees from the Worcester Polytechnic Institute (WPI), Worcester, MA, USA, in 2013 and 2016, respectively, all in electrical engineering.

He was a Post-Doctoral Fellow with the Department of Biomedical Engineering, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA, from 2016 to 2019. He has been an Associate Professor with the Department of Electrical Engineering, Fudan University, Shanghai, China, since February 2019. His research interests include biomedical signal processing, human–machine systems, human rehabilitation, and biometrics.

Edward (Ted) A. Clancy (Senior Member, IEEE) received the B.S. degree from the Worcester Polytechnic Institute (WPI), Worcester, MA, USA, and the M.S. and Ph.D. degrees from the Massachusetts Institute of Technology (MIT), Cambridge, MA, USA, all in electrical engineering.

He has worked in the industry for medical instrumentation and analysis companies interested in electromyogram (EMG), electroencephalogram (EEG), electrocardiogram (ECG), blood pressure, and the defense industry (aircraft instruments and radar). He is currently a Professor of electrical and computer engineering and biomedical engineering with WPI. He is interested in signal processing, stochastic estimation, applied system identification, and instrumentation, particularly as applied to problems in medical engineering and human rehabilitation.

Wei Chen (Senior Member, IEEE) received the B.Eng. and M.Eng. degrees from the School of Electronics and Information Engineering, Xi'an Jiaotong University, Xi'an, China, in 1999 and 2002, respectively, and the Ph.D. degree from the Department of Electrical and Electronics Engineering, The University of Melbourne, Melbourne, VIC, Australia, in 2007.

She took an internship with Bell Laboratories, Alcatel-Lucent, Stuttgart, Germany, in 2005. She was a Research Assistant with the Department of Electrical and Electronics Engineering, The University of Melbourne, in 2007. From 2007 to 2015, she was an Assistant Professor with the Eindhoven University of Technology, Eindhoven, The Netherlands, where she served as the Chair of the Theme Health Care with the Department of Industrial Design from 2009 to 2013. Since October 2015, she has been a Full Professor and the Director of the Center for Intelligent Medical Electronics (CIME), Department of Electronic Engineering, School of Information Science and Technology, Fudan University, Shanghai, China. Her research interests include patient health monitoring, medical monitoring system design using wearable sensors, sleep monitoring, brain activity monitoring, wireless body area networks, ambient intelligence, personalized and smart environment, smart sensor systems, and signal processing.

Dr. Chen is also an Associate Editor of the IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING (TNSRE) and the IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS (JBHI) and the Managing Editor of the IEEE REVIEWS IN BIOMEDICAL ENGINEERING (R-BME).